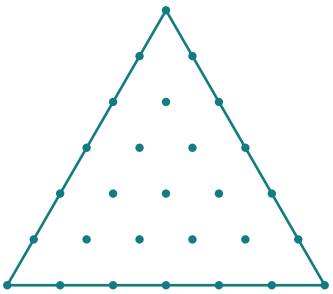
PROBABILITY AND MATHEMATICAL MODELLING

Discrete Information Geometry

Finite Differences, Probability, and Information Flows

Chapter 10



A course in ten chapters

https://bobovski66.github.io/IntroDiscreteInfoGeom/

CHAPTER 10

Graph divergence, diffusion, and information flow

10.1 Information moves on a network

Let $G = (V, E)$ be a finite undirected graph. A probability distribution places mass $p_i(n) \geq 0$ at each vertex, with total mass one. Give each undirected edge a single orientation for bookkeeping and denote the signed current from i to j by $J_{ij}(n) = -J_{ji}(n)$.

The **discrete divergence** at a vertex is net outward current,

$$\operatorname{div} J(i) = \sum_{j \sim i} J_{ij}.$$

Then the conservation law is the discrete continuity equation

$$\boxed{p_i(n+1) - p_i(n) = -\operatorname{div} J_n(i).}$$

Add this equation over every vertex. Each edge current cancels with its opposite, so the total probability remains one. This is a finite analogue of the divergence theorem.

10.2 A graph gradient and Laplacian

For a scalar field f on the vertices, the gradient along an oriented edge $i \to j$ is $f_j - f_i$. A diffusion current flowing from higher to lower probability can be chosen as

$$J_{ij} = \tau(p_i - p_j), \qquad \tau > 0.$$

Substitution yields

$$\boxed{p_i(n+1) = p_i(n) + \tau \sum_{j \sim i} (p_j(n) - p_i(n)).}$$

The sum is a **graph Laplacian** applied to p. It measures how a vertex differs from its neighbors.

To ensure nonnegative probabilities at every step, choose $\tau \leq 1/d_{\max}$, where $d_{\max}$ is the largest vertex degree. Then the update matrix is stochastic. Because the graph is undirected, its rows also sum to one, so it is doubly stochastic.

10.3 Three vertices, one exact step

On the path A–B–C, choose $\tau = 1/4$ and start at $p_0 = (1, 0, 0)$. The step gives

$$p_1 = (3/4, 1/4, 0).$$

The next gives

$$p_2 = (5/8, 5/16, 1/16).$$

Compute Δp_0 and Δp_1. In each case the three coordinates sum to zero, because the changes are redistribution. Repeating the update brings the distribution toward the stationary uniform state.

10.4 Entropy and the direction of change

Because this symmetric diffusion is doubly stochastic, it does not decrease Shannon entropy:

$$H(p_{n+1}) - H(p_n) \geq 0.$$

One way to prove this is to apply the contraction of relative entropy toward the uniform distribution u. Since $D(p\|u) = \log|V| - H(p)$ and $Ku = u$,

$$D(Kp\|u) \leq D(p\|u).$$

Hence $\Delta H_n \geq 0$. The conclusion depends on the **doubly stochastic** assumption. An arbitrary stochastic channel may lower entropy.

10.5 The course's complete mathematical circle

We started with $\Delta a_n = a_{n+1} - a_n$. Now the same operation measures probability changes, Lyapunov dissipation, and entropy production, while the graph gradient and divergence describe where probability travels. The fundamental theorem for finite differences and the cancellation of internal fluxes explain why discrete conservation laws work.

For the final investigation, choose a small graph and initial probability vector, compute at least four iterations, verify mass conservation, graph fluxes, and changes in Shannon entropy, then compare the limiting behavior with your predictions. Use finite calculations to explain what the mathematics says about information moving across the network.

10.6 Further investigations: Graph divergence, diffusion, and information flow

An observation can be recorded at several resolutions. A detailed label may be replaced by a broader group, passed through a noisy channel, or transported among nodes of a

network. These operations transform probability tables by finite weighted sums. They connect graph dynamics with a basic principle of information geometry: applying the same observation rule to two models cannot make them more distinguishable by the discrepancies studied here.

10.7 A stochastic matrix describes one step

Let K_{ji} be the probability of arriving at output label j from input label i. Our convention places the input in the column. Thus

$$K_{ji} \geq 0, \qquad \sum_j K_{ji} = 1 \quad \text{for each } i.$$

Such a table is **column-stochastic**. If input probabilities are p_i, the output probabilities are

$$p_j^+ = \sum_i K_{ji} p_i, \qquad p^+ = Kp.$$

The product symbol Kp abbreviates these weighted sums. Adding over j gives $\sum_j p_j^+ = \sum_i p_i \sum_j K_{ji} = 1$. This is a linear difference equation on probability vectors.

Repeated application gives $p_t = K^t p_0$. A probability vector π is **stationary** if $K\pi = \pi$. Stationarity is a fixed-point condition. It does not by itself imply that every trajectory converges to π.

Example 10.1 (A lazy walk on a three-node path)**.** Consider

$$K = \begin{pmatrix} 3/4 & 1/4 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1/4 & 3/4 \end{pmatrix}.$$

Each column describes remaining at a node or moving to a neighbor. Starting at $p_0 = (1, 0, 0)$ gives $p_1 = (3/4, 1/4, 0)$ and $p_2 = (5/8, 5/16, 1/16)$. The uniform table is stationary because the rows, as well as the columns, sum to one. This particular walk approaches uniform, as can be checked from its repeated finite updates.

A matrix whose rows and columns both sum to one is **doubly stochastic**. The cyclic permutation matrix is also doubly stochastic, but every noncentral three-category state repeats with period three. To establish convergence in a general network, additional conditions are required. Our contraction inequalities below concern one-step comparisons and do not require a claim of convergence.

10.8 Coarse-graining is a deterministic channel

Partition the input labels into groups. The output reports only the group containing the input. Set $K_{ji} = 1$ if label i belongs to group j, and zero otherwise. Each column contains exactly one entry equal to one. The output probabilities are sums over the groups.

Example 10.2 (A distinction erased by grouping)**.** Let $q = (0.4, 0.1, 0.4, 0.1)$ and $p = (0.1, 0.4, 0.1, 0.4)$. Group labels $1, 2$ together and labels $3, 4$ together. Both output distributions are $(1/2, 1/2)$. Before grouping, $D(q\|p) = 0.6 \ln 4 \approx 0.8318$. After grouping, it is zero. The within-group differences cannot be recovered from the recorded group alone.

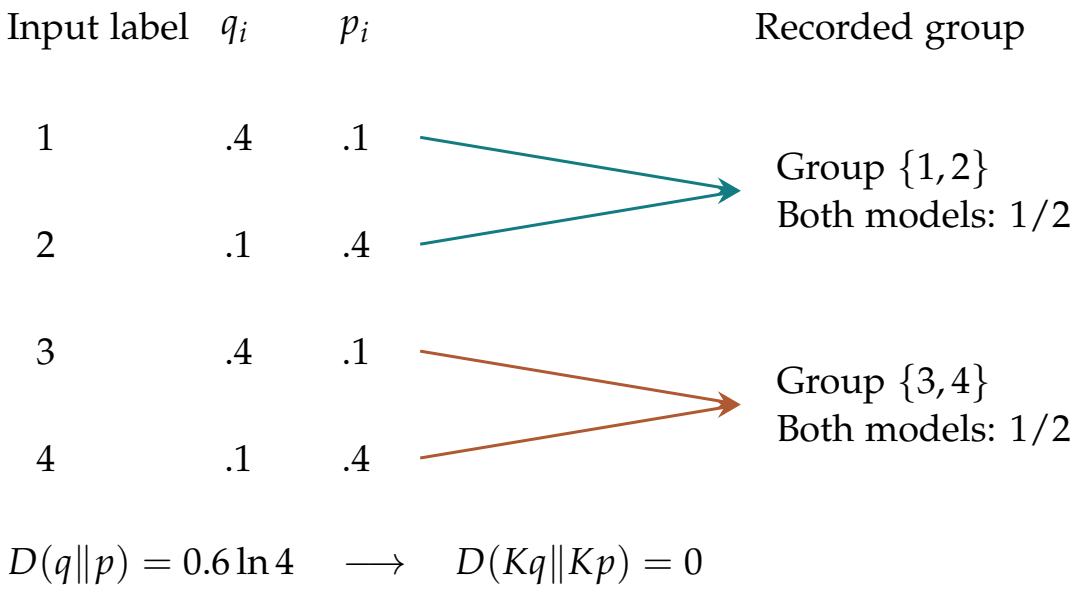


Figure 10.1. Grouping can erase a distinction carried entirely by the probabilities inside each group. Both recorded output tables are identical.

10.9 The log-sum inequality

Let a_i, b_i be nonnegative numbers with positive totals $A = \sum_i a_i$ and $B = \sum_i b_i$, and suppose $b_i > 0$ wherever $a_i > 0$. Applying Gibbs' inequality to a_i/A and b_i/B gives

$$\sum_i a_i \ln \frac{a_i}{b_i} \geq A \ln \frac{A}{B}.$$

To see this directly, multiply $D(a/A\|b/B) \geq 0$ by A and expand the logarithm. This is the **log-sum inequality**. It compares the divergence contributions before and after a set of entries is merged.

Apply it separately to each output label j, using $a_i = K_{ji}q_i$ and $b_i = K_{ji}p_i$. The output totals are $(Kq)_j$ and $(Kp)_j$. Summing over outputs gives

$$D(Kq\|Kp) \leq \sum_{j,i} K_{ji}q_i \ln \frac{q_i}{p_i} = D(q\|p),$$

because each column of K sums to one. Terms with $K_{ji} = 0$ contribute zero. The usual support conventions cover zero totals and infinite divergences.

Proposition 10.1 (Data processing for relative entropy)**.** *For a column-stochastic matrix applied to both distributions,* $D(Kq\|Kp) \le D(q\|p)$.

The same channel must be used for both tables. Applying different recording rules can create differences caused by those rules themselves. For deterministic grouping with positive probabilities, equality holds exactly when q_i/p_i is constant within each group: no distinction relevant to that ratio is lost within a group.

10.10 Square-root overlap also contracts distance

For each output label, Cauchy–Schwarz gives

$$\sqrt{\left(\sum_i K_{ji}q_i\right)\left(\sum_i K_{ji}p_i\right)} \ge \sum_i K_{ji}\sqrt{q_i p_i}.$$

Adding over j and using column sums yields

$$B(Kq, Kp) \ge B(q, p).$$

Since inverse cosine decreases as its argument increases, the spherical distance satisfies

$$d(Kq, Kp) \le d(q, p).$$

Hellinger distance also cannot increase, because its square is $1 - B$. Thus two different constructions, a logarithmic discrepancy and a square-root distance, both obey an information-loss principle under the same finite observation rule.

Example 10.3 (A noisy binary observation)**.** Suppose a binary label is flipped with probability $1/4$, so $K = \left(\begin{smallmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{smallmatrix}\right)$. The tables $(0.9, 0.1)$ and $(0.1, 0.9)$ become $(0.7, 0.3)$ and $(0.3, 0.7)$. Their overlap increases from 0.6 to $2\sqrt{0.21} \approx 0.9165$. Their spherical distance decreases from approximately 1.8546 to 0.8230.

10.11 When does entropy increase on a network?

A stochastic channel can lower entropy. A channel that sends every input to one output converts every distribution into a certain outcome with entropy zero. The data-processing inequality compares two models through the same channel; entropy measures the uncertainty of one table. The questions are related but distinct.

For a doubly stochastic matrix, uniform u is stationary. Data processing then gives

$$D(Kp\|u) = D(Kp\|Ku) \le D(p\|u).$$

Using $D(p\|u) = \ln k - H(p)$ shows $H(Kp) \ge H(p)$. This is an exact entropy-increase statement with a specific assumption on the channel. Equality can occur at a nonuniform table, as a permutation simply rearranges its probabilities.

10.12 A finite statistical family

A collection of probability tables indexed by one or more parameters is a **statistical family**. On fixed labels x_i, for example, the weights

$$p_i(m, a) = \frac{e^{-a(x_i-m)^2}}{\sum_j e^{-a(x_j-m)^2}}, \qquad a > 0,$$

form a finite bell-shaped family. Changing m shifts the emphasis across the labels, and changing a changes its concentration. These are normalized weights on a fixed finite set. Their entropy, relative entropy, and spherical distance can all be computed by the formulas already developed. Grouping labels produces a coarser family and contracts pairwise discrepancies by the same theorems.

This offers a capstone investigation: choose two parameter values, calculate their tables, apply a recording channel, and quantify the distinctions retained. The finite sums, token graphs, and exact updates developed throughout the course provide the tools for carrying out the investigation and explaining its assumptions.

10.13 Solving the three-node walk exactly

The particular walk introduced earlier can be solved by tracking two contrasts. Write

$$a_t = p_{1,t} - p_{3,t}, \qquad b_t = p_{1,t} - 2p_{2,t} + p_{3,t}.$$

Substituting the three update equations and collecting coefficients gives

$$a_{t+1} = \frac{3}{4}a_t, \qquad b_{t+1} = \frac{1}{4}b_t.$$

Both are geometric sequences. Combining these contrasts with $p_{1,t} + p_{2,t} + p_{3,t} = 1$ recovers the probabilities:

$$p_{1,t} = \frac{1}{3} + \frac{b_t}{6} + \frac{a_t}{2}, \quad p_{2,t} = \frac{1}{3} - \frac{b_t}{3}, \quad p_{3,t} = \frac{1}{3} + \frac{b_t}{6} - \frac{a_t}{2}.$$

Because $(3/4)^t$ and $(1/4)^t$ approach zero, every initial table for this walk approaches uniform. This proves the claim for the stated matrix by two scalar recurrences, without needing a general matrix convergence theorem.

Example 10.4 (An exact three-step distribution)**.** From $(1, 0, 0)$, both initial contrasts are one. At step three, $a_3 = 27/64$ and $b_3 = 1/64$. The recovered table is $(35/64, 21/64, 8/64)$. Direct multiplication of the step-two table $(5/8, 5/16, 1/16)$ by the walk matrix gives the same answer.

10.14 Where the lost divergence goes under grouping

For deterministic groups G_j, define output masses $Q_j = \sum_{i \in G_j} q_i$ and $P_j = \sum_{i \in G_j} p_i$. Assume positive input tables. Within group j, form conditional tables q_i/Q_j and p_i/P_j. Splitting the logarithm gives

$$D(q\|p) = D(Q\|P) + \sum_j Q_j D\left(q(\cdot \mid G_j)\middle\|p(\cdot \mid G_j)\right).$$

The first term is the discrepancy retained in the group labels. The second is the average discrepancy among labels inside a group. Every lost contribution is nonnegative. Equality under grouping occurs exactly when the two within-group conditional tables agree, equivalent to a constant ratio q_i/p_i in each group.

Example 10.5 (Retained and lost distinctions)**.** For the four-label tables $(0.4, 0.1, 0.4, 0.1)$ and $(0.1, 0.4, 0.1, 0.4)$, adjacent-pair grouping gives the same output masses. All the divergence is in the conditional tables $(0.8, 0.2)$ and $(0.2, 0.8)$ within each pair. Grouping labels $1, 3$ together instead gives outputs $(0.8, 0.2)$ and $(0.2, 0.8)$, while each within-group conditional table is uniform. This second grouping preserves the full divergence.

10.15 Composing observation rules

If channel K is followed by channel L, the combined rule is the matrix product LK. Its entries are $(LK)_{hi} = \sum_j L_{hj} K_{ji}$. Each term represents passing through intermediate output j. Nonnegativity is immediate, and adding over h gives one for every input column. Thus the composition is again a stochastic channel.

Applying data processing twice gives

$$D(LKq\|LKp) \le D(Kq\|Kp) \le D(q\|p).$$

The spherical distances obey the same succession of inequalities. This gives a mathematical description of a recording pipeline: sampling labels, adding noise, grouping outputs, and recording an even coarser category can be analyzed one finite operation at a time. A stage may preserve a particular pair's discrepancy, but a later common stage cannot restore a distinction already erased from those output tables.

10.16 Exercises

10.1 Verify the column sums of the three-node walk matrix and compute one update of $(1/2, 1/4, 1/4)$.

10.2 Show algebraically that a column-stochastic matrix maps probability vectors to probability vectors.

10.3 Find the stationary table of the binary flip channel in the example.

10.4 Write the matrix for the cyclic permutation $(p_A, p_B, p_C) \mapsto (p_C, p_A, p_B)$ and verify its period.

10.5 For $q = (0.4, 0.1, 0.4, 0.1)$ and $p = (0.1, 0.4, 0.1, 0.4)$, compute their overlap before and after the stated grouping.

10.6 Derive the log-sum inequality from Gibbs' inequality.

10.7 Give a grouping for which two distinct positive tables retain their relative entropy exactly. Explain the ratio condition within groups.

10.8 Verify the relative-entropy contraction numerically for the binary flip example.

10.9 Prove that a doubly stochastic matrix preserves the uniform distribution.

10.10 Give a stochastic channel that strictly lowers the entropy of the uniform distribution. Explain why this does not violate data processing.

10.11 On labels $-2, -1, 0, 1, 2$, calculate the finite bell-shaped table with $m = 0$ and $a = \ln 2$. Compare its entropy with uniform entropy.

10.12 Design a capstone comparing two probability models before and after grouping. Include a data table, both directed relative entropies, spherical distance, a channel specification, and an interpretation of any lost distinction.

10.17 Selected solutions

1. The new table is $(7/16, 5/16, 1/4)$.

3. Solving $x = (3/4)x + (1/4)(1 - x)$ gives $x = 1/2$, so the stationary table is uniform.

5. The input overlap is $4\sqrt{0.04} = 0.8$. The equal outputs have overlap one.

7. Group labels $1, 2$ together and $3, 4$ together. Take $p = (0.1, 0.2, 0.3, 0.4)$ and $q = (0.2, 0.4, 6/35, 8/35)$. Both sum to one, and the within-group ratios are 2 and $4/7$. Log-sum has equality in each group, so the divergence is preserved.

9. For $u_i = 1/k$, the output entry is $(Ku)_j = (1/k)\sum_i K_{ji} = 1/k$ by the row-sum condition.

11. The unnormalized weights are $(1/16, 1/2, 1, 1/2, 1/16)$ with total $17/8$. The table is $(1/34, 4/17, 8/17, 4/17, 1/34)$. It is nonuniform and has entropy strictly less than $\ln 5$.

Laboratory investigation

1. On A–B–C with $p = (1, 0, 0)$ and $\tau = 1/4$, (a) compute both signed edge currents J_{AB}, J_{BC}, (b) calculate outward divergence at A, B, C, (c) use $\Delta p = -\operatorname{div} J$ to determine the next state, and (d) check mass conservation.

2. Using the same path, calculate p_1 through p_4. (a) Write each state as a probability vector. (b) Compute all coordinate differences. (c) Verify every difference vector sums to zero. (d) Explain the approach toward uniform.
3. For $p_0 = (1, 0, 0)$ and the first three transport updates, (a) compute entropy at each step, (b) calculate its first differences, (c) identify the sign of every difference, and (d) explain which property of the diffusion matrix supports the pattern.
4. Compare two channels: the symmetric path diffusion and a channel sending every input to A. (a) Construct each stochastic matrix. (b) Calculate both output distributions from uniform input. (c) Compare input/output entropy. (d) Explain why a general stochastic channel does not have to raise entropy.
5. Capstone: choose a four- or five-node undirected graph and an initial probability distribution. (a) Define a legal diffusion step using $\tau \leq 1/d_{\max}$. (b) Calculate at least four iterates with a table or short program. (c) Show currents, divergence, mass conservation, and entropy changes. (d) Compare two states by Fisher–Rao distance and explain what the mathematical results say about the flow of information.